

PROBLEM–SOLUTION FIT

Strategic Product Placement Analysis

Problem Overview

Retail sales performance is influenced by product category, product position, consumer demographics, foot traffic, pricing, promotions, and seasonal trends. When these factors are viewed as raw data or separate reports, comparing them and identifying meaningful product placement patterns becomes difficult. Retail users need a clearer way to connect these factors with sales performance and support informed placement decisions.

IDENTIFIED PROBLEM	PROJECT SOLUTION	FIT
Product category performance is difficult to compare quickly.	Create category-wise average sales volume visualizations.	Makes strong and weak category performance easier to identify.
The effect of store position is not clear in raw data.	Compare aisle, end-cap, and front-of-store sales performance.	Supports evaluation of product placement effectiveness.
Customer groups are difficult to analyze across sales results.	Visualize sales volume by consumer demographics.	Highlights differences in customer-related sales patterns.
Foot traffic and sales relationships are difficult to interpret.	Compare foot traffic with average sales volume.	Provides a visual view of traffic-related sales patterns.
Product and competitor prices require direct comparison.	Visualize product price and competitor price together.	Supports easier pricing comparison.
Promotion and seasonal effects may be missed.	Analyze promotional and seasonal sales patterns.	Adds business context to product performance.
Static reports limit exploration.	Build an interactive Tableau dashboard with filters.	Allows users to explore retail factors dynamically.
Important findings may be disconnected.	Create a Tableau story with structured scenes.	Presents key findings in a clear analytical sequence.
Analytics need a clear presentation interface.	Embed Tableau views in a Flask and Bootstrap website.	Provides a structured and accessible project experience.

FIT ANALYSIS

Why the Proposed Solution Matches the Problem

CLEAR VISUAL COMPARISON	The dashboard converts multiple retail measures into charts that are easier to compare than raw records.
INTERACTIVE EXPLORATION	Filters allow users to study categories, positions, demographics, and other factors based on analysis needs.
CONNECTED RETAIL INSIGHTS	Sales volume is analyzed together with placement, traffic, pricing, promotion, season, and customer information.
STRUCTURED STORYTELLING	The Tableau story presents selected findings in sequence instead of leaving insights scattered across separate views.
WEB ACCESSIBILITY	Flask and Bootstrap provide a clean interface for viewing the embedded dashboard and story.
DECISION SUPPORT	The final analysis helps users recognize patterns that can support product placement and retail planning decisions.

Problem–Solution Fit Statement

Strategic Product Placement Analysis fits the identified retail problem by transforming complex and connected sales factors into interactive visual insights.

The Tableau dashboard supports comparison and exploration, the Tableau story organizes important findings, and the Flask web interface provides a clear presentation layer. Together, these components directly address the difficulty of interpreting product placement and sales data through static or disconnected information.

Design Outcome

The problem–solution fit analysis confirms that the selected project design directly responds to the identified retail analytics challenges. The solution focuses on clear comparisons, interactive exploration, structured visual storytelling, and web-based presentation, making it suitable for strategic product placement analysis.

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